

newtest

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```
# Load libraries
library(ggplot2)
library(car)

## Warning: package 'car' was built under R version 4.3.3
## Loading required package: carData
library(caret)

## Loading required package: lattice
library(MASS)
library(knitr)
library(broom)

## Warning: package 'broom' was built under R version 4.3.3
library(kableExtra)

data = read.csv("StudentPerformanceFactors.csv")
#multiple linear regression
full_model = lm(Exam_Score ~ Hours_Studied + Sleep_Hours + Physical_Activity +
  factor(Internet_Access) + Tutoring_Sessions + Attendance +
  factor(Motivation_Level) +
  factor(Parental_Involvement) + factor(Family_Income) +
  factor(Parental_Education_Level) +
  factor(Teacher_Quality) + factor(School_Type) +
  factor(Peer_Influence) + factor(Distance_from_Home) +
  factor(Gender) + factor(Learning_Disabilities)+
  factor(Extracurricular_Activities)+factor(Access_to_Resources),
  data = data)

# Tidy the results for cleaner output
model_summary = tidy(full_model)
kable(model_summary, caption = "Table 1 Regression Model Summary: Exam Score Predictors")
```

Table 1: Table 1 Regression Model Summary: Exam Score Predictors

term	estimate	std.error	statistic	p.value
(Intercept)	45.1480221	0.5069695	89.0547098	0.0000000
Hours_Studied	0.2978464	0.0044288	67.2526752	0.0000000
Sleep_Hours	-0.0124897	0.0180733	-0.6910564	0.4895545
Physical_Activity	0.1786696	0.0257919	6.9273588	0.0000000

term	estimate	std.error	statistic	p.value
factor(Internet_Access)Yes	0.9395090	0.1004315	9.3547284	0.0000000
Tutoring_Sessions	0.4893883	0.0215527	22.7066003	0.0000000
Attendance	0.1975218	0.0023009	85.8446101	0.0000000
factor(Motivation_Level)Low	-1.0764916	0.0770165	-13.9774135	0.0000000
factor(Motivation_Level)Medium	-0.5837399	0.0701265	-8.3240973	0.0000000
factor(Parental_Involvement)Low	-1.9447979	0.0770032	-25.2560775	0.0000000
factor(Parental_Involvement)Medium	-1.0493375	0.0618347	-16.9700287	0.0000000
factor(Family_Income)Low	-1.0485264	0.0735418	-14.2575541	0.0000000
factor(Family_Income)Medium	-0.5559786	0.0736135	-7.5526707	0.0000000
factor(Parental_Education_Level)College	0.3943796	0.2326319	1.6952943	0.0900669
factor(Parental_Education_Level)High School	-0.0574361	0.2306725	-0.2489941	0.8033732
factor(Parental_Education_Level)Postgraduate	0.9115282	0.2352397	3.8748918	0.0001077
factor(Teacher_Quality)High	0.4994901	0.2492037	2.0043447	0.0450741
factor(Teacher_Quality)Low	-0.5114413	0.2585926	-1.9777879	0.0479943
factor(Teacher_Quality)Medium	-0.0763490	0.2467496	-0.3094188	0.7570128
factor(School_Type)Public	0.0498020	0.0577091	0.8629841	0.3881777
factor(Peer_Influence)Neutral	0.5281838	0.0719781	7.3381162	0.0000000
factor(Peer_Influence)Positive	0.9985253	0.0716967	13.9270824	0.0000000
factor(Distance_from_Home)Far	-0.4341034	0.2766421	-1.5691878	0.1166523
factor(Distance_from_Home)Moderate	-0.0537918	0.2680890	-0.2006491	0.8409791
factor(Distance_from_Home)Near	0.4823394	0.2659549	1.8136136	0.0697828
factor(Gender)Male	-0.0392004	0.0536980	-0.7300173	0.4654056
factor(Learning_Disabilities)Yes	-0.8391339	0.0864883	-9.7022785	0.0000000
factor(Extracurricular_Activities)Yes	0.5673705	0.0541088	10.4857282	0.0000000
factor(Access_to_Resources)Low	-2.1075333	0.0768212	-27.4342743	0.0000000
factor(Access_to_Resources)Medium	-1.0121318	0.0613440	-16.4992884	0.0000000

```
#Variable Selection
```

```
# Perform stepwise AIC
```

```
selected_model = stepAIC(full_model, direction = "both")
```

```
## Start: AIC=10165.41
```

```
## Exam_Score ~ Hours_Studied + Sleep_Hours + Physical_Activity +
```

```
##   factor(Internet_Access) + Tutoring_Sessions + Attendance +
```

```
##   factor(Motivation_Level) + factor(Parental_Involvement) +
```

```
##   factor(Family_Income) + factor(Parental_Education_Level) +
```

```
##   factor(Teacher_Quality) + factor(School_Type) + factor(Peer_Influence) +
```

```
##   factor(Distance_from_Home) + factor(Gender) + factor(Learning_Disabilities) +
```

```
##   factor(Extracurricular_Activities) + factor(Access_to_Resources)
```

```
##
```

```
##           Df Sum of Sq  RSS   AIC
```

```
## - Sleep_Hours      1      2 30499 10164
```

```
## - factor(Gender)    1      2 30500 10164
```

```
## - factor(School_Type) 1      3 30501 10164
```

```
## <none>                30497 10165
```

```
## - Physical_Activity  1     223 30720 10211
```

```
## - factor(Internet_Access) 1     406 30903 10251
```

```
## - factor(Learning_Disabilities) 1     436 30934 10257
```

```
## - factor(Extracurricular_Activities) 1     510 31007 10273
```

```
## - factor(Teacher_Quality) 3     662 31159 10301
```

```
## - factor(Distance_from_Home) 3     699 31196 10309
```

```
## - factor(Parental_Education_Level) 3     917 31414 10355
```

```

## - factor(Motivation_Level)          2      911 31408 10356
## - factor(Peer_Influence)            2      925 31422 10359
## - factor(Family_Income)             2      981 31478 10370
## - Tutoring_Sessions                 1     2391 32888 10662
## - factor(Parental_Involvement)       2     3059 33556 10793
## - factor(Access_to_Resources)        2     3538 34035 10887
## - Hours_Studied                     1    20972 51469 13621
## - Attendance                        1    34171 64668 15130
##
## Step: AIC=10163.89
## Exam_Score ~ Hours_Studied + Physical_Activity + factor(Internet_Access) +
##   Tutoring_Sessions + Attendance + factor(Motivation_Level) +
##   factor(Parental_Involvement) + factor(Family_Income) + factor(Parental_Education_Level) +
##   factor(Teacher_Quality) + factor(School_Type) + factor(Peer_Influence) +
##   factor(Distance_from_Home) + factor(Gender) + factor(Learning_Disabilities) +
##   factor(Extracurricular_Activities) + factor(Access_to_Resources)
##
##                                     Df Sum of Sq  RSS   AIC
## - factor(Gender)                   1         2 30502 10162
## - factor(School_Type)               1         3 30503 10163
## <none>                             30499 10164
## + Sleep_Hours                      1         2 30497 10165
## - Physical_Activity                 1        223 30722 10210
## - factor(Internet_Access)           1        405 30904 10249
## - factor(Learning_Disabilities)     1        438 30937 10256
## - factor(Extracurricular_Activities) 1        510 31009 10271
## - factor(Teacher_Quality)           3        661 31161 10300
## - factor(Distance_from_Home)        3        699 31198 10308
## - factor(Parental_Education_Level)  3        916 31416 10354
## - factor(Motivation_Level)          2        911 31410 10354
## - factor(Peer_Influence)            2        927 31426 10358
## - factor(Family_Income)             2        983 31482 10369
## - Tutoring_Sessions                 1     2393 32892 10661
## - factor(Parental_Involvement)       2     3060 33560 10792
## - factor(Access_to_Resources)        2     3542 34041 10886
## - Hours_Studied                     1    20970 51469 13619
## - Attendance                        1    34189 64689 15130
##
## Step: AIC=10162.42
## Exam_Score ~ Hours_Studied + Physical_Activity + factor(Internet_Access) +
##   Tutoring_Sessions + Attendance + factor(Motivation_Level) +
##   factor(Parental_Involvement) + factor(Family_Income) + factor(Parental_Education_Level) +
##   factor(Teacher_Quality) + factor(School_Type) + factor(Peer_Influence) +
##   factor(Distance_from_Home) + factor(Learning_Disabilities) +
##   factor(Extracurricular_Activities) + factor(Access_to_Resources)
##
##                                     Df Sum of Sq  RSS   AIC
## - factor(School_Type)               1         3 30505 10161
## <none>                             30502 10162
## + factor(Gender)                   1         2 30499 10164
## + Sleep_Hours                      1         2 30500 10164
## - Physical_Activity                 1        222 30724 10208
## - factor(Internet_Access)           1        404 30906 10247
## - factor(Learning_Disabilities)     1        437 30938 10254

```

```
## - factor(Extracurricular_Activities) 1      509 31011 10270
## - factor(Teacher_Quality)           3      662 31164 10298
## - factor(Distance_from_Home)        3      699 31201 10306
## - factor(Parental_Education_Level)  3      917 31418 10352
## - factor(Motivation_Level)          2      910 31412 10353
## - factor(Peer_Influence)            2      926 31428 10356
## - factor(Family_Income)             2      983 31485 10368
## - Tutoring_Sessions                 1     2394 32896 10660
## - factor(Parental_Involvement)      2     3064 33566 10791
## - factor(Access_to_Resources)       2     3542 34043 10884
## - Hours_Studied                     1    20973 51475 13618
## - Attendance                        1    34187 64689 15128
##
## Step: AIC=10161.17
## Exam_Score ~ Hours_Studied + Physical_Activity + factor(Internet_Access) +
##   Tutoring_Sessions + Attendance + factor(Motivation_Level) +
##   factor(Parental_Involvement) + factor(Family_Income) + factor(Parental_Education_Level) +
##   factor(Teacher_Quality) + factor(Peer_Influence) + factor(Distance_from_Home) +
##   factor(Learning_Disabilities) + factor(Extracurricular_Activities) +
##   factor(Access_to_Resources)
##
##               Df Sum of Sq  RSS   AIC
## <none>                        30505 10161
## + factor(School_Type)         1         3 30502 10162
## + factor(Gender)              1         2 30503 10163
## + Sleep_Hours                 1         2 30503 10163
## - Physical_Activity           1        222 30727 10207
## - factor(Internet_Access)     1        405 30910 10246
## - factor(Learning_Disabilities) 1        437 30942 10253
## - factor(Extracurricular_Activities) 1        510 31015 10269
## - factor(Teacher_Quality)     3        661 31166 10297
## - factor(Distance_from_Home)  3        699 31204 10305
## - factor(Parental_Education_Level) 3        915 31420 10350
## - factor(Motivation_Level)    2        911 31417 10352
## - factor(Peer_Influence)      2        927 31432 10355
## - factor(Family_Income)       2        985 31490 10367
## - Tutoring_Sessions           1       2396 32901 10659
## - factor(Parental_Involvement) 2       3062 33567 10789
## - factor(Access_to_Resources) 2       3538 34044 10882
## - Hours_Studied               1    20973 51478 13616
## - Attendance                  1    34186 64691 15126

# Summary of the selected model
summary(selected_model)
```

```
##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Physical_Activity +
##   factor(Internet_Access) + Tutoring_Sessions + Attendance +
##   factor(Motivation_Level) + factor(Parental_Involvement) +
##   factor(Family_Income) + factor(Parental_Education_Level) +
##   factor(Teacher_Quality) + factor(Peer_Influence) + factor(Distance_from_Home) +
##   factor(Learning_Disabilities) + factor(Extracurricular_Activities) +
##   factor(Access_to_Resources), data = data)
##
```

```

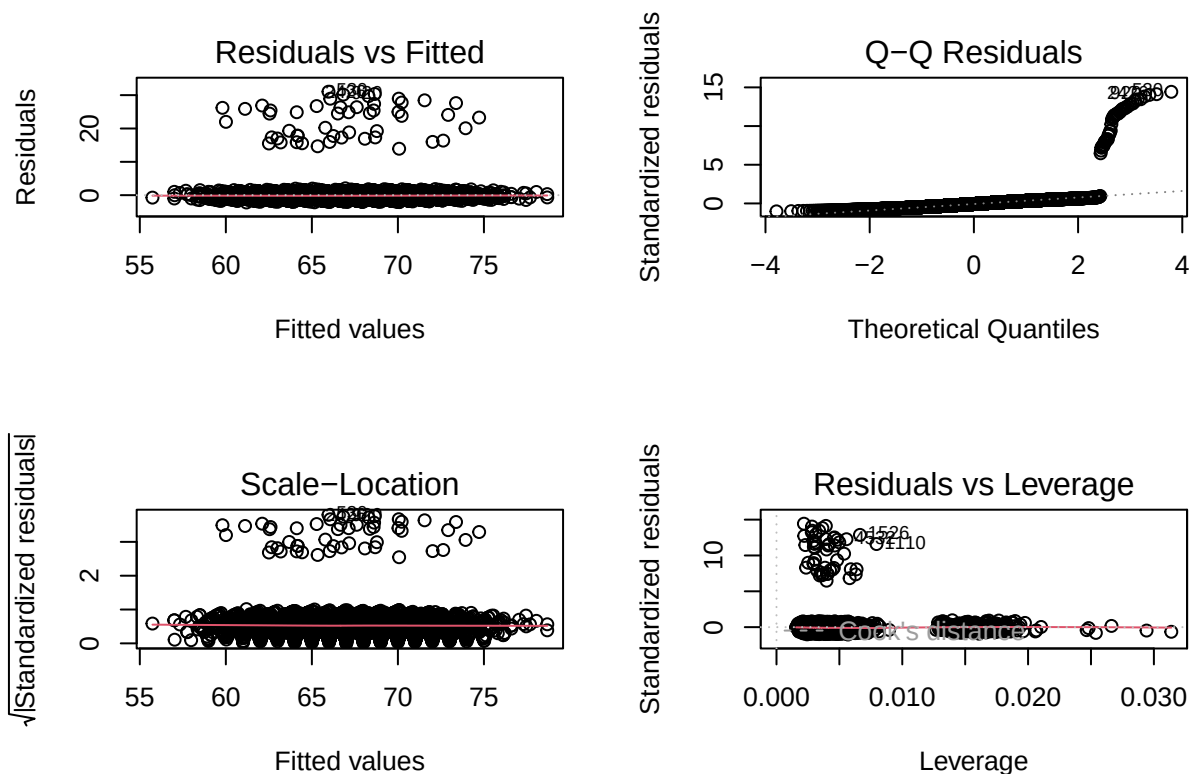
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1979 -0.7704 -0.1616  0.4477 31.0325
##
## Coefficients:
##                                Estimate Std. Error t value
## (Intercept)                   45.087250   0.490297  91.959
## Hours_Studied                   0.297829   0.004428  67.260
## Physical_Activity               0.178330   0.025787   6.915
## factor(Internet_Access)Yes      0.938711   0.100396   9.350
## Tutoring_Sessions              0.489832   0.021548  22.732
## Attendance                     0.197500   0.002300  85.871
## factor(Motivation_Level)Low     -1.076560   0.076998 -13.982
## factor(Motivation_Level)Medium -0.584005   0.070118  -8.329
## factor(Parental_Involvement)Low -1.944793   0.076973 -25.266
## factor(Parental_Involvement)Medium -1.050579   0.061811 -16.997
## factor(Family_Income)Low        -1.050478   0.073514 -14.290
## factor(Family_Income)Medium     -0.557152   0.073598  -7.570
## factor(Parental_Education_Level)College 0.389622   0.232526   1.676
## factor(Parental_Education_Level)High School -0.062430   0.230573  -0.271
## factor(Parental_Education_Level)Postgraduate 0.904925   0.235098   3.849
## factor(Teacher_Quality)High      0.495098   0.249134   1.987
## factor(Teacher_Quality)Low       -0.515101   0.258516  -1.993
## factor(Teacher_Quality)Medium    -0.080500   0.246677  -0.326
## factor(Peer_Influence)Neutral     0.526715   0.071957   7.320
## factor(Peer_Influence)Positive    0.998826   0.071677  13.935
## factor(Distance_from_Home)Far     -0.433761   0.276555  -1.568
## factor(Distance_from_Home)Moderate -0.054948   0.267986  -0.205
## factor(Distance_from_Home)Near     0.481601   0.265861   1.811
## factor(Learning_Disabilities)Yes  -0.839168   0.086460  -9.706
## factor(Extracurricular_Activities)Yes 0.567289   0.054101  10.486
## factor(Access_to_Resources)Low     -2.107146   0.076791 -27.440
## factor(Access_to_Resources)Medium  -1.010626   0.061308 -16.484
##                                Pr(>|t|)
## (Intercept)                   < 2e-16 ***
## Hours_Studied                   < 2e-16 ***
## Physical_Activity               5.10e-12 ***
## factor(Internet_Access)Yes      < 2e-16 ***
## Tutoring_Sessions              < 2e-16 ***
## Attendance                     < 2e-16 ***
## factor(Motivation_Level)Low     < 2e-16 ***
## factor(Motivation_Level)Medium < 2e-16 ***
## factor(Parental_Involvement)Low < 2e-16 ***
## factor(Parental_Involvement)Medium < 2e-16 ***
## factor(Family_Income)Low        < 2e-16 ***
## factor(Family_Income)Medium     4.24e-14 ***
## factor(Parental_Education_Level)College 0.09386 .
## factor(Parental_Education_Level)High School 0.78658
## factor(Parental_Education_Level)Postgraduate 0.00012 ***
## factor(Teacher_Quality)High      0.04693 *
## factor(Teacher_Quality)Low       0.04635 *
## factor(Teacher_Quality)Medium    0.74418
## factor(Peer_Influence)Neutral    2.78e-13 ***
## factor(Peer_Influence)Positive   < 2e-16 ***

```

```
## factor(Distance_from_Home)Far          0.11683
## factor(Distance_from_Home)Moderate      0.83755
## factor(Distance_from_Home)Near          0.07011 .
## factor(Learning_Disabilities)Yes        < 2e-16 ***
## factor(Extracurricular_Activities)Yes   < 2e-16 ***
## factor(Access_to_Resources)Low          < 2e-16 ***
## factor(Access_to_Resources)Medium      < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.153 on 6580 degrees of freedom
## Multiple R-squared:  0.6949, Adjusted R-squared:  0.6937
## F-statistic: 576.4 on 26 and 6580 DF, p-value: < 2.2e-16
```

```
# Residual diagnostics
```

```
par(mfrow = c(2, 2))
plot(selected_model)
```



```
# Check for multicollinearity using Variance Inflation Factor (VIF)
```

```
vif(selected_model)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## Hours_Studied    1.002659  1      1.001329
## Physical_Activity 1.007660  1      1.003823
## factor(Internet_Access) 1.002962  1      1.001480
## Tutoring_Sessions 1.001863  1      1.000931
## Attendance       1.005080  1      1.002537
## factor(Motivation_Level) 1.008482  2      1.002114
## factor(Parental_Involvement) 1.007059  2      1.001760
## factor(Family_Income) 1.007923  2      1.001975
```

```
## factor(Parental_Education_Level) 1.010683 3 1.001773
## factor(Teacher_Quality) 1.010507 3 1.001744
## factor(Peer_Influence) 1.007396 2 1.001844
## factor(Distance_from_Home) 1.010678 3 1.001772
## factor(Learning_Disabilities) 1.002754 1 1.001376
## factor(Extracurricular_Activities) 1.004355 1 1.002175
## factor(Access_to_Resources) 1.008364 2 1.002085

# Residual vs Fitted Plot for final model
ggplot(data = data.frame(Fitted = fitted(selected_model), Residuals = residuals(selected_model)),
      aes(x = Fitted, y = Residuals)) +
  geom_point(color = "blue", alpha = 0.6) +
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
  labs(title = "Table 2 Residuals vs Fitted Values", x = "Fitted Values", y = "Residuals") +
  theme_minimal()
```

Table 2 Residuals vs Fitted Values

